INCH-POUND
MIL-PRF-8805 /74K
31 May 2001
SUPERSEDING
MIL-PRF-8805 /74J
12 May 1995

PERFORMANCE SPECIFICATION SHEET

# SWITCHES, SENSITIVE, SPDT, 4 AMPERES, HERMETIC SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-8805

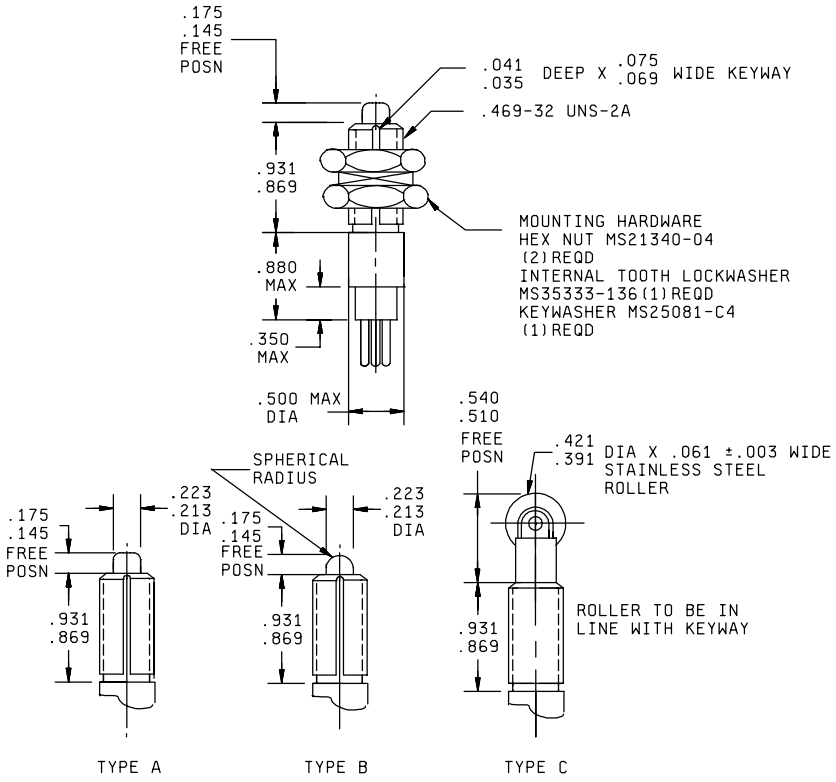


FIGURE 1. Dimensions and configuration


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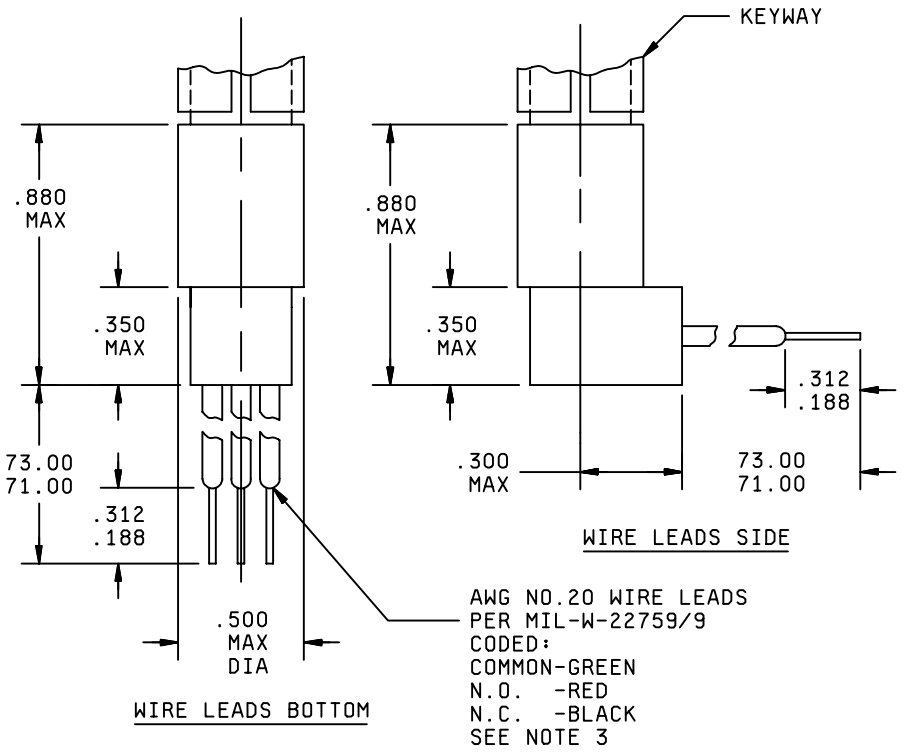


| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| 0.003 | .0762 | 0.213 | 5.410 | 0.510 | 12.954 |
| 0.035 | .8890 | 0.223 | 5.664 | 0.540 | 13.716 |
| 0.041 | 1.041 | 0.300 | 7.620 | 0.869 | 22.073 |
| 0.061 | 1.549 | 0.312 | 7.924 | 0.880 | 22.352 |
| 0.069 | 1.752 | 0.350 | 8.890 | 0.931 | 23.647 |
| 0.075 | 1.905 | 0.391 | 9.931 | 71.00 | 1803.4 |
| 0.145 | 3.683 | 0.421 | 10.693 | 73.00 | 1854.2 |
| 0.175 | 4.445 | 0.469 | 11.913 | | |
| 0.188 | 4.775 | 0.500 | 12.700 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Exact shape of switch is optional within dimensions specified.
4. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for lockwasher. Dimensions shall be in accordance with the referenced hardware.

FIGURE 1. Dimensions and configurations – continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 5 (hermetic). Basic switch shall be hermetically sealed M8805/68-022 listed on QPL-8805. Resilient plunger seal.

Temperature characteristic: 2 (- 65°C to + 125°C).

Shock type: M (100 g, test condition I, method 213 of MIL-STD-202).

Vibration grade: 3 (10 Hz to 3,000 Hz swept sinusoidal ).

Operating characteristics: see table II.

Actuating force: 6.5 pounds maximum.

Releasing force: 1.5 pound minimum.

Pretravel: .020 inch (0.51 mm) maximum

Movement differential: .005 inch (0.13 mm) maximum.

Overtravel: .125 inch (3.18 mm) minimum.

Contact resistance: .04 ohm + 0.011 ohm per lead per foot.

After electrical endurance: In accordance with MIL-PRF-8805, plus 0.011 ohm per lead per foot.

Dielectric withstanding voltage:

Sea level and 70,000 feet: 1,050 V rms (terminal to case), 800 V rms (terminal to terminal), 500 V rms after electrical life.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 25,000 cycles

Intermediate current: 25,000 cycles.

Low level: 25,000 cycles.

Electrical ratings: see table III.

Short circuit: 20 times rated resistive load (SAE-AS33201; PIN MS3320-7 ½, or equivalent used for circuit protection)..

Weight: 1.0 ounce plus 0.1 ounce per lead per foot.

Qualification:

Group submission: See table IV.

Military Part or Identifying Number (PIN): See tables I and V.



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Table 1. PIN;s and configuration.

| PIN 1/ | Resistive load rating | Plunger seal | Type of actuating mechanism | Terminals |
|---|---|---|---|---|
| M8805/74-035 | 4 Amps | None | A | Wire leads – bottom |
| M8805/74-036 | 4 Amps | | C | Wire leads – bottom |
| M8805/74-037 | 4 Amps | | A | Wire leads – side |
| M8805/74-038 | 4 Amps | | C | Wire leads – side |
| M8805/74-039 | 4 Amps | Resilient Seal | A | Wire leads – bottom |
| M8805/74-040 | 4 Amps | | C | Wire leads – bottom |
| M8805/74-041 | 4 Amps | | A | Wire leads – side |
| M8805/74-042 | 4 Amps | | C | Wire leads – side |
| M8805/74-043 | 4 Amps | None | D | Wire leads – bottom |
| M8805/74-044 | 4 Amps | | D | Wire leads – side |
| M8805/74-045 | 4 Amps | Resilient Seal | D | Wire leads – bottom |
| M8805/74-046 | 4 Amps | | D | Wire leads – side |

1/ PIN's M8805/74-001 through –034 have been canceled. See Table V for substitution information..

Table II. Operating characteristics.

| Plunger seal | Actuating Force (maximum) | Releasing Force (minimum) | Movement Differential (maximum) | Pretravel Inches (maximum) | Overtravel (minimum) |
|---|---|---|---|---|---|
| Unsealed | 30 ounces 1/ | 4 ounces 1/ | .005 inch (0.13 mm) | .010 (0.25 mm) | .125 inch (3.18 mm) |
| Resilient | 7.0 pounds | 1.5 pounds | .005 inch (0.13 mm) | .020 (0.51 mm) | .125 inch (3.18 mm) |

1/ At sea level and 70,000 feet.

Table III. Electrical ratings.

| Load | Sea level and 70,000 feet (28 V dc) |
|---|---|
| | (amperes) |
| Resistive | 4 |
| Inductive | 1 |
| Lamp | 2 |

Table IV. Qualification.

| Inspection | Basic Switch | Other switch samples | Extent of approval |
|---|---|---|---|
| Qualification inspection table of MIL-PRF-8805, groups I through VI, inclusive | M8805/74-040 (all sample units) | M8805/74- 035 through –039, -041 through -046 (2 sample units each): Visual and mechanical inspection only | All |



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Table V. Substitution information for canceled PIN's. 1/

| Canceled PIN's M8805/74- | | Substitute PIN's M8805/74- |
|---|---|---|
| 004 | 023 | 035 |
| 006 | 024 | 036 |
| 007 | 025 | 037 |
| 009 | 026 | 038 |
| 013 | 027 | 039 |
| 015 | 028 | 040 |
| 016 | 029 | 041 |
| 018 | 030 | 042 |
| 019 | 031 | 043 |
| 020 | 032 | 044 |
| 021 | 033 | 045 |
| 022 | 034 | 046 |

1/ The canceled PIN's are no longer procurable. Substitute, PIN configurations are interchangeable and are rated at 4 amperes for 25,000 cycles..

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army – AR, AV, MI
Navy – AS, MC, OS, SH
Air force – 19, 99

Preparing activity:
DLA - CC

(Project 5930-1671-03)



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